

Electromagnetic Theory (EE3005)

Sessional-I Exam

Date: September 20, 2025

Course Instructor(s)

1. Dr. Afzal Ahmed (Course Moderator)
2. Mohsin Yousuf

Total Time (Hrs): 1

Total Marks: 40

Total Questions: 2

Roll No

Section

Student Signature

Do not write below this line

1. Attempt all questions and remember to solve parts of the same question together.
2. Final answers should be correct up to two decimal places with proper SI units.
3. Show all the steps with the help of diagrams and equations.

CLO # 01: Demonstrate the use of 3D orthogonal coordinate system and vector analysis tools in problem solving

Q1: A vector field is given in rectangular coordinate system (RCS) as:

[20 marks]

$$\mathbf{Q}_R = \frac{\sqrt{x^2 + y^2 + z^2}}{\sqrt{x^2 + y^2}} [(x - y)\mathbf{a}_x + (x + y)\mathbf{a}_y]$$

Zunaira likes eating ice-cream cone, help her in answering the following questions using Figure 1.

- (a) Convert the above vector field $\mathbf{Q}_R$ in spherical coordinate system (SCS) and name it $\mathbf{Q}_S$. [7]
- (b) Compute the line integral, $\int_L \mathbf{Q}_S \cdot d\mathbf{L}$ where, L is the circular edge of the ice-cream cone by specifying the $d\mathbf{L}$ element first. [3]
- (c) Compute the surface integral $\int_{S_1} \mathbf{Q}_S \cdot d\mathbf{S}_1$ where, S_1 is the top surface. Also, find $\int_{S_2} \mathbf{Q}_S \cdot d\mathbf{S}_2$ where, S_2 is the slanting conical surface of the shape. [6]
- (d) Determine the total volume of the ice-cream cone using dv in SCS. [4]

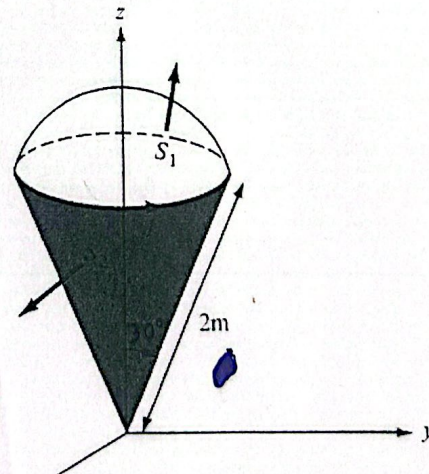


Figure 1. Volume in the form of ice cream cone

Hint Q1:

$$d\mathbf{L}_{SCS} = dr \mathbf{a}_r + r d\theta \mathbf{a}_\theta + r \sin \theta d\phi \mathbf{a}_\phi$$

$$d\mathbf{S}_r = r^2 \sin \theta d\theta d\phi \mathbf{a}_r, d\mathbf{S}_\theta = r \sin \theta dr d\phi \mathbf{a}_\theta, d\mathbf{S}_\phi = r dr d\theta \mathbf{a}_\phi, dv = r^2 \sin \theta dr d\theta d\phi$$

National University of Computer and Emerging Sciences
Lahore Campus

$\begin{aligned} x &= r \sin \theta \cos \phi \\ y &= r \sin \theta \sin \phi \\ z &= r \cos \theta \end{aligned}$	$\begin{aligned} r &= \sqrt{x^2 + y^2 + z^2}, (r \geq 0) \\ \theta &= \cos^{-1} \frac{z}{\sqrt{x^2 + y^2 + z^2}}, (0^\circ \leq \theta \leq 180^\circ) \\ \phi &= \tan^{-1} \frac{y}{x} \end{aligned}$
--	--

	$\mathbf{a}_r$	$\mathbf{a}_\theta$	$\mathbf{a}_\phi$
$\mathbf{a}_x$	$\sin \theta \cos \phi$	$\cos \theta \cos \phi$	$-\sin \phi$
$\mathbf{a}_y$	$\sin \theta \sin \phi$	$\cos \theta \sin \phi$	$\cos \phi$
$\mathbf{a}_z$	$\cos \theta$	$-\sin \theta$	0

CLO # 02: Analyze electrostatic fields and/or its properties governed by Coulomb's / Gauss's law for a given charge distribution in free space and / or dielectrics.

Q2: **[20 marks]**

- (a) A uniform volume charge density, $\rho_v = 0.2 \text{ } [\mu\text{C}/\text{m}^3]$ is present throughout the spherical shell extending from $r = 3 \text{ cm}$ to $r = 5 \text{ cm}$. It is given that $\rho_v = 0$ elsewhere. **[8]**

- (i) Find the total electric flux leaving the shell Ψ_{shell} .
- (ii) Also find r_1 if half of the total charge computed in part (i) is located in the region: $3 < r < r_1 \text{ [cm]}$.

Hint Q2(a): The total electric flux leaving the shell is the amount of charge present in the shell.

- (b) Three infinite sheets of charge are located in free space as follows: **[12]**

1. $\rho_{S1} = 10 \text{ nC}/\text{m}^2$ at $y = -4$
2. $\rho_{S2} = 6 \text{ nC}/\text{m}^2$ at $y = 1$
3. $\rho_{S3} = -10 \text{ nC}/\text{m}^2$ at $y = 4$.

Answer the following questions:

- (i) Draw the charge distribution in RCS.
- (ii) Find electric field intensity E everywhere.